

pandoc + xelatex PDF export environment setup

Documents the install steps and background for the priority 1 (pandoc + xelatex) PDF export option referenced in SKILL.md, so the environment can be set up in one pass on a new machine.

Why this chain

Compared to weasyprint / wkhtmltopdf / Chrome headless / md-to-pdf:

- **Best typography** —LaTeX is a publication-grade engine; pagination, widow/orphan control, letter spacing, tables of contents, footnotes, and cross-references are all a level more refined than the HTML→PDF path
- **Most reliable for Chinese** —driving real system fonts directly via xelatex + CJKmainfont="PingFang SC" avoids tofu boxes, dropped characters, and incorrect punctuation compression
- **Code blocks / tables / formulas** —syntax highlighting, alignment, and math formulas are all native LaTeX quality
- **Consistent across machines** —PDFs produced by LaTeX aren't affected by browser version / DPI / font fallback
- **Good for long documents** —automatic tables of contents, chapter numbering, and headers/footers are first-class citizens

Trade-off: larger footprint (BasicTeX ~100MB / MacTeX ~5GB), slower on first install.

macOS installation (BasicTeX route recommended)

```
# 1. pandoc (skip if already installed)
brew install pandoc

# 2. BasicTeX (a slimmed-down TeX Live, ~100MB)
brew install --cask basictex

# 3. Refresh PATH (BasicTeX installs to /Library/TeX/texbin)
eval "$(usr/libexec/path_helper)"
# To persist this: append the following line to ~/.zshrc
# export PATH="/Library/TeX/texbin:$PATH"

# 4. Upgrade tlmgr and install the packages needed for Chinese PDFs
sudo tlmgr update --self
sudo tlmgr install xecjk ctex fandol collection-fontsrecommended
```

If you have plenty of disk space and want to do it all in one go (a complete font set, no further tlmgr installs needed), install this instead:

```
brew install --cask mactex-no-gui    # ~4GB, no GUI tools, but full command-line toolchain
```

Verify

```
which pandoc xelatex
pandoc --version | head -1
xelatex --version | head -1
```

Both commands printing a path and version number means you're ready.

Producing a PDF

Full command for documents with Chinese text + ASCII topology diagrams (the typical shape of network-topology docs):

```
pandoc input.md -o "TITLE.pdf" --pdf-engine=xelatex \
  -V CJKmainfont="PingFang SC" -V mainfont="PingFang SC" \
  -V monofont="Menlo" -V CJKmonofont="PingFang SC" \
```

```
-V geometry:margin=1.5cm \
-V monofontoptions="Scale=0.80" -V CJKmonofontoptions="Scale=0.80" \
2>warnings.log; echo "exit=$?"; grep -c "Missing character" warnings.log
```

All three font settings are required: CJKmainfont covers body Chinese text; monofont (Menlo) covers box-drawing characters `—|_|` inside code blocks; CJKmonofont covers Chinese labels inside code blocks. Setting only CJKmainfont leaves ASCII diagrams broken and Chinese text dropped inside code blocks —because code blocks use a separate monospace font that CJKmainfont doesn't reach.

After exporting, `grep -c "Missing character" warnings.log` must be 0: xelatex drops missing characters silently, with exit code still 0 —checking only the exit code will miss broken diagrams.

The fonts used, PingFang SC / Menlo, are both built into macOS —no extra install needed.

Common pitfalls

xelatex: command not found (still not found after installing)

/Library/TeX/texbin isn't on PATH. To fix it for the current shell temporarily:

```
eval "$( /usr/libexec/path_helper )"
```

To persist it: append the following line to `~/.zshrc` (or `~/.bashrc`):

```
export PATH="/Library/TeX/texbin:$PATH"
```

! LaTeX Error: File 'xeCJK.sty' not found.

BasicTeX doesn't ship xeCJK by default; run once:

```
sudo tlmgr install xecjk ctex
```

Chinese characters showing as boxes / dropped

`-V CJKmainfont=...` wasn't passed, or the font name passed isn't installed on the system. Built-in macOS options include:

- PingFang SC (recommended default)
- Heiti SC
- Songti SC

On non-macOS systems, use Noto Sans CJK SC instead (needs to be installed separately).

ASCII diagrams broken / Chinese text dropped inside code blocks

Produces a flood of `Missing character: There is no (U+2500) ... in font [lmmono10-regular]` (or the same warning for Chinese characters). Root cause: code blocks use a separate monospace font, and the default Immono10 has neither box-drawing characters nor Chinese glyphs —`-V CJKmainfont` doesn't reach it, since it only covers body text. Add both font settings:

```
-V monofont="Menlo" -V CJKmonofont="PingFang SC"
```

Wide diagrams (70+ column topology diagrams) will also overflow the page width and get truncated; add `-V geometry:margin=1.5cm + -V monofontoptions="Scale=0.80" -V CJKmonofontoptions="Scale=0.80"`.

Colored emoji (etc.) show up blank in the PDF

xelatex doesn't support colored emoji fonts (Apple Color Emoji is a bitmap font, and forcing it usually errors out or still renders blank). If emoji appear in key content, switch to SKILL.md's priority 4 (Chrome headless); if emoji only appear in descriptive prose, it's acceptable to drop them.

Running pandoc prints a bunch of `command not found: _encode / _decode`

Errors that look like:

```
setValueForKeyFakeAssocArray:27: command not found: _encode  
valueForKeyFakeAssocArray:28: command not found: _decode
```

Come from the combination of `zsh + eval "$(/usr/libexec/path_helper)"`, triggered when some associative-array helper functions aren't auto-loaded in a non-interactive shell. Doesn't affect pandoc —exit is still 0 and the PDF is still produced.

To get rid of it, skip `path_helper` and export explicitly instead:

```
export PATH="/Library/TeX/texbin:$PATH"
```

Once that line is in `~/.zshrc`, pandoc invocations in new shells will be clean.

tlmgr: Local TeX Live ... is older than remote repository

```
sudo tlmgr update --self --all
```

If it still errors (usually happens across a major-version year boundary), consult the official TeX Live upgrade guide to manually upgrade the distribution version.

Related

- SKILL.md —this skill's main document; the 5-tier priority list for the PDF export toolchain is in its "PDF export toolchain" section